



Data Science & Artificial Intelligence

Machine Learning



Decision Tree

DPP-01

Discussion Notes



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[MCQ]

#Q. Consider a binary classification problem with the following dataset

Sample	Feature 1	Feature 2	Class label
A	2	3	0 ✓
B	1	5	1 ✓
C	4	2	1 ✓
D	3	1	0 ✓
E	5	4	0 ✓
F	2	2	1 ✓
G	3	3	1 ✓

Sol. Entire data

0: 3

1: 4

$P_0 = 3/7$ $P_1 = 4/7$

$G_I = 1 - P_0^2 - P_1^2 = 0.489$

Calculate the Gini impurity for the entire dataset



0.514



0.486



0.353



0.65

[MCQ]

#Q. We will use the dataset below to learn a decision tree which predicts if people pass machine learning (Yes or No), based on their previous GPA (High, Medium, or Low) and whether or not they studied.

GPA	Studied	Passed
L	F	F ✓
L	T	T
M	F	F ✓
M	T	T
H	F	T
H	T	T

Sol

$$H(P) \Rightarrow \begin{matrix} F: 2 & P_F = 1/3 \\ T: 4 & P_T = 2/3 \end{matrix}$$

↓

$$\frac{1}{3} \log_2 3 + \frac{2}{3} \log_2 3/2 = 0.918$$

What is the entropy $H(\text{Passed})$?



0.46



0.63



0.92



0.23

[MCQ]



#Q. In a binary classification problem, a classifier correctly predicts 90 instances as positive, incorrectly predicts 15 instances as positive when they are negative, correctly predicts 90 instances as negative, and incorrectly predicts 10 instances as negative when they are positive. What is the accuracy of the classifier?

A 80

B 83

C 88 ✓

D 93

		Actual	
		P	N
Pred	P	90	15
	N	10	90

$$\begin{aligned}\text{Accuracy} &= \frac{TP + TN}{\text{Total}} \times 100 \\ &= \frac{90 + 90}{205} \times 100 \\ &= 87.8\%\end{aligned}$$

#Q. Which of the following statements is/are true about Gini Impurity in Decision trees?

- ☒ **A** Gini index is a measure of impurity or purity used while creating a decision tree in the CART
- ☐ **B** ~~X~~ Gini Impurity is the measurement of changes in entropy after the segmentation of a dataset based on an attribute
- ☐ **C** ~~X~~ An attribute with a high Gini index should be preferred as compared to the low Gini index
- ☐ **D** ~~X~~ It only creates binary splits, and the CART algorithm uses the Gini index to create binary splits.

[MCQ]

#Q. Suppose you are working on a binary classification problem, and you decide to use logistic regression to model the relationship between a single feature (X) and the binary outcome (Y). You collect a dataset and fit the logistic regression model: $P(Y = 1|X) = \frac{1}{1+e^{(\beta_0+\beta_1X)}}$

After training, you obtain the following logistic regression coefficients:

$$\beta_0 = -3.2$$

$$\beta_1 = 0.8$$

Now, you want to make predictions and evaluate the model's performance

For a given observation with $X=2.5$, calculate the predicted probability of $Y = 1$, and predict the class



0.99, class 1



0.23, class 1



0.23, class:0



0.77, class:0

Sol Simply Put Value
in formula
 $P_1 = 0.99 \rightarrow \text{class 1}$

[MCQ]

#Q. Consider a dataset with the following distribution of classes for a binary classification problem:

Class A: 40 instances

Class B: 60 instances

After the first split based on a certain feature, the dataset is divided into two subsets:

Subset 1: Class A: 30 instances

Class B: 10 instances

Subset 2: Class A: 10 instances

Class B: 50 instances

Determine the information gain achieved by the split

Sol Parent $P_A = 0.4$ $P_B = 0.6$

$E = 0.970$

Child: Subset 1 $P_A = 3/4$ $P_B = 1/4 \Rightarrow$

$E_{S1} = 0.811$



0.787



0.389



0.307



0.26

$$\text{Subset 2} \quad P_A = 1/6 \quad P_B = 5/6$$

$$E = 0.650$$

$$E_c = \frac{60 \times 0.650 + 40 \times 0.811}{100} = 0.7144$$

$$IG = E_p - E_c = \underline{\underline{0.26}}$$

[MCQ]

#Q. Assume that you are given a data set and a neural network model trained on the data set. You are asked to build a decision tree model with the sole purpose of understanding/interpreting the built neural network model. In such a scenario, which among the following measures would you concentrate most on optimising?

- A** Accuracy of the decision tree model on the given data set
- B** F1 measure of the decision tree model on the given data set
- C** Fidelity of the decision tree model, which is the fraction of instances on which the neural network and the decision tree give the same output
- D** Comprehensibility of the decision tree model, measured in terms of the size of the corresponding rule set

- Here we want DT to mimic or to perform similar to NN
- So we want similar performance

[NAT]

#Q. In a decision tree, if a node has 3 possible outcomes with probabilities 0.4, 0.3 and 0.3 what is the entropy? (Rounded up to 2 decimal)

$$E_p = 0.4 \log_2 \frac{1}{0.4} + 0.3 \log_2 \frac{1}{0.3} + 0.3 \log_2 \frac{1}{0.3}$$

$$= \underline{\underline{1.5709}}$$

[MCQ]

#Q. Consider the following statements:

✓ S_1 : Decision Trees are prone to overfitting, especially when they are allowed to grow deep without pruning.

S_2 : Cross-Validation provides an unbiased estimate of model performance by using multiple training and validation splits.

Which of the above is true?

A Only S_1

B Only S_2

✓ **C** Both S_1 and S_2

D None of these

[MCQ]

#Q. Consider the following statements:

S_1 : A Decision Tree is a generative model. ✗

S_2 : Cross-Validation is used to evaluate the performance of a model on unseen data.

Which of the above is true?

A Only S_1

B Only S_2

C Both S_1 and S_2

D None of these

[MSQ]

#Q. Which of the following criteria is used in the top-down decision tree approach to evaluate the best split at a node?

decision tree we use Gini Index
IG

- A** Gini Index ✓✓
- B** Mean Squared Error (MSE)
- C** Information Gain ✓✓
- D** Kappa Statistic

[MCQ]

#Q. Consider the following statements:

- ☒ S_1 : The depth of a decision tree directly controls its bias and variance.
- ☐ S_2 : Decision trees are naturally resistant to overfitting.
- ☒ S_3 : Entropy in decision trees is a measure of randomness in the data.
- ☐ S_4 : In decision trees, features are chosen for splitting based on their correlation with the output variable.

Which of the above statements is correct?



Only S_1 and S_2



Only S_2 and S_3



Only S_1 and S_3 ☒



Only S_2 and S_4

[MCQ]

#Q. Consider a node in a decision tree that contains 80 samples. Out of these, 50 samples belong to Class X, 20 samples belong to Class Y and 10 samples belong to Class Z. Calculate the Gini Index for this node.

$$\begin{aligned}
 G_i I &= 1 - (p_x^2 + p_y^2 + p_z^2) \\
 &= 1 - \left(\frac{5}{8}\right)^2 - \left(\frac{2}{8}\right)^2 - \left(\frac{1}{8}\right)^2 \Rightarrow 0.531
 \end{aligned}$$

A 0.32

B 0.46

C 0.53 ✓

D 0.38

[MCQ]

#Q. A decision tree classifier learned from a fixed training set achieves 100% accuracy. Which of the following models trained using the same training set will also achieve 100% accuracy?

- (i) Logistic regressor. ✗
- (ii) A polynomial of degree one kernel SVM. ✗
- (iii) A linear discriminant function. ✗
- (iv) Naïve Bayes classifier. ✗

A

i

B

i and ii

C

All of the above

D



None of the above

• decision tree = 100% accuracy

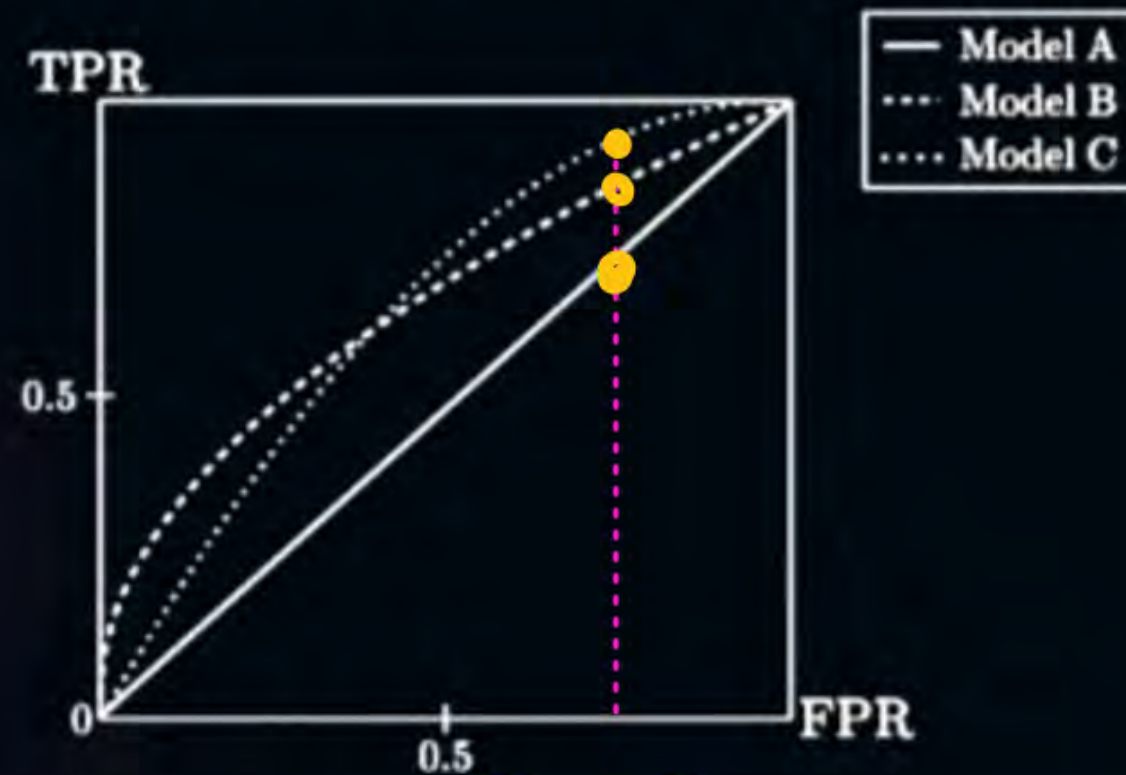


overfit

• So we want to search algo that do overfitting

[MCQ]

#Q. Suppose we fix the decision threshold for all 3 logistic regression models to be such that we get a FPR of 0.8 when we evaluate our model. In order of most to least preferred, rank each of the models given their ROC curves.



• So $FPR = 0.8$ so
we select algo that
give higher TPR
Model $C > B > A$

A $C > B > A$ ✓

B $C > A > B$

C $B > C > A$

D $B > A > C$

[MCQ]

#Q. Consider training a decision tree given a design matrix

$X = \begin{bmatrix} 6 & 3 \\ 2 & 7 \\ 9 & 6 \\ 4 & 2 \end{bmatrix}$ and labels $y = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$. Let f_1 denote feature 1, corresponding to the first column of X , and let f_2 denote feature 2, corresponding to the second column. Which of the following splits at the root node gives the highest information gain?

a) $f_1 > 2 \rightarrow \begin{matrix} 6, 9, 4 \rightarrow 1, 1, 0 \\ 2 \rightarrow 0 \end{matrix}$

A $f_1 > 2$

B $f_2 > 3$

C $f_1 > 4$

D $f_2 > 6$

Sol we need min Entropy of child

b) $f_2 > 3 \rightarrow \begin{matrix} 2, 3 & 1, 0 \\ 6, 7 & 1, 0 \end{matrix}$

[MCQ]

#Q. Consider training a decision tree given a design matrix

$X = \begin{bmatrix} 6 & 3 \\ 2 & 7 \\ 9 & 6 \\ 4 & 2 \end{bmatrix}$ and labels $y = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$. Let f_1 denote feature 1, corresponding to the first column of X , and let f_2 denote feature 2, corresponding to the second column. Which of the following splits at the root node gives the highest information gain?

A $f_1 > 2$

B $f_2 > 3$

C $f_1 > 4$ ✓

D $f_2 > 6$

c) $f_1 > 4 \rightarrow \begin{matrix} 6, 9 = 1, 1 \\ \rightarrow 2, 4 = 0, 0 \end{matrix} \rightarrow \text{min } \epsilon''$

Sol we need min Entropy of child

d) $f_2 > 6$ $\begin{matrix} 7 & 0 \\ 2, 3, 6 & 1, 1, 0 \end{matrix}$

#Q. Details of ten international cricket games between two teams "Green" and "Blue" are given in Table C. This table consists of matches played on different pitches, across formats along with their winners. The attribute Pitch can take one of two values: spin-friendly (represented as S or pace-friendly (represented as F. The attribute Format can take one of two values: one-day match (represented as O) or test match (represented as T).

A cricket organization would like to use the information given in Table C to develop a decision-tree model to predict outcomes of future games between these two teams.

To develop such a model, the computed Information Gain(C, Pitch) with respect to the Target is (rounded off to two decimal places).

done in class

Match Number	Pitch	Format	Winner (Target)
1	S	T	Green
2	S	T	Blue
3	F	O	Blue
4	S	O	Blue
5	F	T	Green
6	F	O	Blue
7	S	O	Green
8	F	T	Blue
9	F	O	Blue
10	S	O	Green

[MCQ]

#Q. Which of the following is true for a decision tree?

- A. A decision tree is an example of a linear classifier. ✗
- B. The entropy of a node typically decreases as we go down a decision tree. ✓✓
- C. Entropy is a measure of purity. ✗
- D. ✗ An attribute with lower mutual information should be preferred to other attributes.

"Information Gain" we choose attribute with max MI or IG

A A

B B ✓

C C

D D

[MCQ]

#Q. Consider the dataset, S given below:

Elevation	Road Type	Speed Limit	Speed
Steep	Uneven	Yes	Slow
Steep	Smooth	Yes	Slow
Flat	Uneven	No	Fast
Steep	Smooth	No	Fast

→ label

$$H(S) \Rightarrow P_S = \frac{1}{2}$$

$$P_F = \frac{1}{2}$$

$$\underline{\underline{E = 1}}$$

Elevation, Road Type and speed Limit are the features and Speed is the target label that we want to predict.

Find the entropy of the dataset, S as given above:

- a. 0.5
- b. 0
- c. 1
- d. 0.7



a



b



c

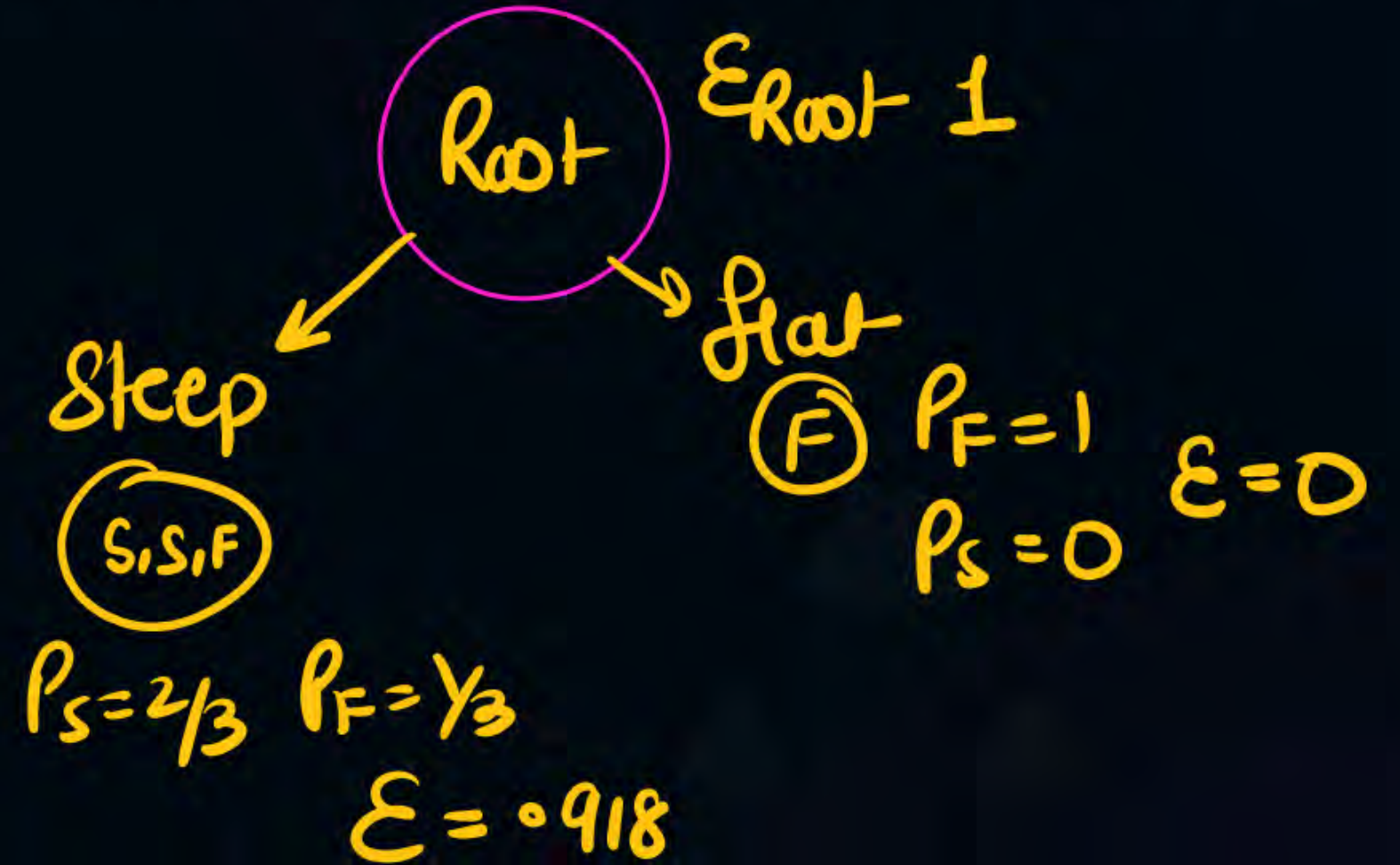


d

[MCQ]

#Q. Find the information Gain if the dataset is split at the feature "Elevation":

- a. 1
- b. 0
- c. 0.675
- d. ~~0.325~~ 0.311



A

a

B

b

C

c

D

d ✓

$$E_C = \frac{0.918 \times 3 + 1 \times 0}{4}$$

$$= 0.688$$

$$IG = E_P - E_C = 0.311$$

[MCQ]

#Q. You have generated data from a 3-degree polynomial with some noise. What do you expect of the model that was trained on this data using a 5- degree polynomial as function class?

- ☒ a. Low bias, high variance
- ☐ b. High bias, low variance
- ☐ c. Low bias, low variance
- ☐ d. High bias, low variance

A

a

*overfitted***B**

b

C

c

D

d

[MCQ]

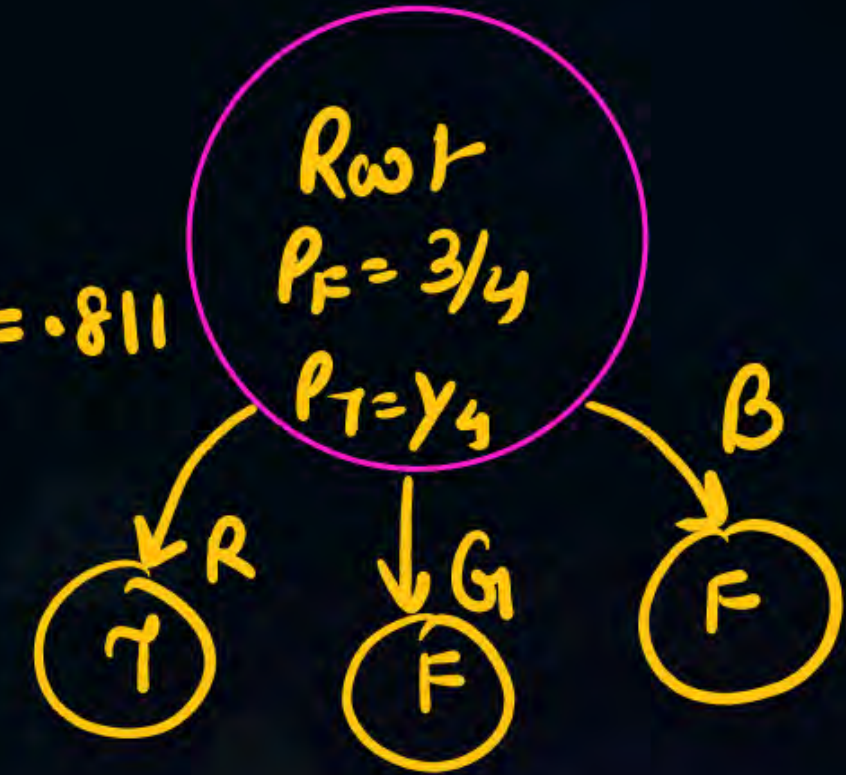
#Q. Given the following table of observations, calculate the information gain $IG(Y|X)$ that would result from learning the value of X

X	Y
Red	True
Green	False
Brown	False
Brown	False

Information Gain wnt

$$IG = E_R - E_C = 0.811 - 0 = 0.811$$

$$E_R = 0.811$$



100% homogeneous
 $E = 0$



1/2



0.811



3/2



None of the above

[MCQ]

#Q. When choosing one feature from $X_1, \dots, X_n$ while building a Decision Tree, which of the following criteria is the most appropriate to maximize? (Here, $H()$ means entropy, and $P()$ means probability)

A $P(Y|X_j)$

B $P(Y) - P(Y|X_j)$

C $H(Y) - H(Y|X_j)$ ✓

D $H(YX_j)$

E $H(Y) - P(Y)$

max I_G

$$= H(Y) - H(Y/x)$$

y is label
 x is dimension



THANK - YOU